SOEN 341 Software Process
Team Project - Fall 2026
Project Title: CareerConnect - Job Search and Application Tracking Platform

Objective

This project will provide students with hands-on experience in software development using Agile practices, GitHub-based collaboration, project management, testing, and continuous integration. The project is divided into four sprints. Teams are encouraged to use Generative AI tools responsibly to support software engineering activities such as requirements elicitation, documentation, planning, testing, and implementation. All AI-generated content must be validated and documented by the team.

Project Description

CareerConnect is a web-based platform designed to help job seekers manage their job search activities. The system allows users to create profiles, upload and manage resumes, search for job opportunities, track submitted applications, and follow the progress of their application process. The platform aims to centralize job-search activities and help users stay organized throughout their career development journey.

Primary Users: **Job Seekers and Recruiters.**

System Features

- User registration, authentication, and profile management.
- Resume upload and management.
- Job posting management for recruiters.
- Job search and filtering capabilities.
- Job application submission.
- Application status tracking (Applied, Interview, Offered, Rejected).
- Application history dashboard.
- Notifications and reminders for application deadlines.
- Saved jobs and favourites.
- AI-assisted resume feedback or job matching suggestions (optional advanced feature).
- Teams are encouraged to design creative and meaningful AI-based functionality that enhances the application.

- ***Note: Note:*** *The project must include at least* ***one Generative AI feature****. This can be either the AI-assisted resume feedback/job-matching feature listed above or another team-proposed feature. In addition, teams must implement* ***one additional original feature****, which may also leverage Generative AI. Students may use an external AI API or run an AI model locally to perform the required AI processing.*

Sprint 1 Delivery Instructions

The goal of Sprint 1 is to understand the system, establish the development environment, organize the team, identify project requirements, and implement the first basic functionalities. The work completed during the sprint will be monitored during the weekly lab meetings with the TA.

Activities	Sprint 1 Details
GitHub Setup and Initialization	Create repository, add team members, define permissions, create project board, organize folders for documentation, minutes, AI logs and sprint deliverables.
README File	Project description, identified problem, proposed solution, team members, technologies, setup instructions, and proposed features.
Sprint Planning	Initial sprint plan including backlog, priorities, risks, effort estimation, and team capacity.
User Stories and Task Breakdown	Minimum 15 user stories using GitHub Issues and associated tasks assigned to team members (Usage of labels).
Team Process Definition	Define workflow, branching strategy, pull request process, code review process, Definition of Ready, and Definition of Done.
AI Usage Log	Document AI tools used, prompts, generated outputs, validation activities, and final decisions. Teams must be prepared to discuss AI usage during lab reviews.
Meeting Minutes	Maintain meeting minutes documenting attendance, decisions, action items, and sprint discussions.
Code Demonstration	Implement and demonstrate at least TWO basic features. Recommended features: (1) User Registration/Login and (2) Resume Upload or User Profile Management.

AI Usage

You may use AI at different stages of the project, such as requirements elicitation, documentation, planning, testing, debugging, and implementation. However, all AI-generated content must be reviewed, validated, and documented by the team. You are not required to use AI for all project activities or deliverables.

AI Usage Log Requirements

Create a directory named AI_Log in your project repository. Inside this directory, each team member must create a subdirectory using their own name.

For each sprint, every team member must upload a **PDF report** to their personal subdirectory documenting all instances in which they used AI during the project.

What Must Be Logged

Every use of AI must be documented, regardless of its purpose or stage in the project. This includes, but is not limited to:

- Requirements analysis and refinement
- Brainstorming and idea generation
- Project planning and task allocation
- System design and architecture discussions
- Code generation or code review
- Debugging and problem solving
- Test case generation
- Documentation writing and editing
- UI/UX design suggestions
- Research and information gathering

If AI influenced a decision, recommendation, design choice, implementation, or deliverable in any way, the interaction must be recorded in the log.

Evidence of AI Interaction

For each logged activity, provide either:

- A link to the AI conversation (if available), **or**
- The complete prompt and relevant AI response included in an appendix.

Required Log Entry Format

For every AI-assisted activity, include the following information:
Task ID/Title: A short description of the task.
Purpose of AI Use: Why AI was used (e.g., brainstorming, requirements analysis, coding, debugging, testing, documentation, review).
Chat Link or Prompt/Response: Link to the conversation or the full prompt and AI response.
AI-Suggested Content: Summary of what the AI proposed or generated.
Validation: How the output was verified (e.g., manual review, testing, peer review, static analysis, CI/CD checks, comparison with requirements).
Decision: Indicate whether the AI output was:

- Accepted
- Modified before use
- Rejected

Reflection: Briefly describe what was learned from the interaction and whether the AI assistance was useful.
Responsible Person: Team member who conducted the interaction and completed the task.

Example Log Entry

Task 1.1: Define User Stories

Purpose of AI Use: Brainstorming and requirements elicitation

Chat Link or Prompt/Response: Chat link provided (or attached in appendix)

AI-Suggested Content: Generated an initial list of user stories for job seekers and recruiters.

Validation: Compared the suggestions against project requirements and reviewed them during the team meeting.

Decision: Modified and partially accepted.

Reflection: AI helped identify missing recruiter requirements, but several generated stories were too generic and needed refinement.

Responsible Person: Jane Smith

Original Team Features and User Stories

As one of your AI Log entries in Sprint 1, prompt an AI assistant to generate **10 user stories** for your project. Log this activity exactly as described in the AI Usage section, including the full prompt (or chat link) and the user stories returned by the AI.

After reviewing the AI-generated stories, the team must conduct its own brainstorming session and propose **additional original user stories and features that were not suggested by the AI**. These team-generated ideas should reflect the team's creativity, understanding of user needs, and vision for improving the application.

Document these ideas in a clearly separate section titled **"Team-Generated User Stories and Features"**. AI-generated and team-generated content must never be mixed or presented as having the same source.

During the lab, each team will be expected to:

- Clearly identify which user stories originated from AI and which were created by the team.
- Highlight and explain all team-generated user stories and feature ideas.
- Justify why these additions improve the application and how they address user needs.
- Demonstrate the team's contribution beyond the AI-generated suggestions.

The quality and originality of the team-generated ideas will be considered when evaluating the project. Teams are expected to extend the AI output rather than simply adopt it.

Sprint 1 Submission Instructions

Submit **one text document** that includes the following:

1. **Cover Page**
    - Project title
    - Names of all group members
2. **README File**
    - A copy of your README file
3. **GitHub Repository Link**
    - Include a working link to your GitHub repository in the README section.

Please ensure that the document is clearly organized, professionally formatted, and that the GitHub link is accessible to the instructor. This submission will be used to verify group membership, project documentation, and repository setup for Sprint 1.

Sprint 1 Lab Monitoring

Teams will meet weekly with their TA. Students must demonstrate repository activity, sprint planning progress, meeting minutes, commits, pull requests, and implementation work. Teams must also present evidence of GenAI usage including prompts, generated outputs, modifications, and validation steps. Students should be able to explain both technical decisions and AI-assisted decisions.

Sprint 1 Rubric

Category	Weight (/15)
Team Members and Repository Setup	0.5
README File	0.5
Feature Identification and Scope Definition	1.0
User Stories	3.0
Sprint Planning	3.0
Task Breakdown	1.5
AI Usage Log and Validation Evidence	2.0
Detailed Log of Each Team Member Contribution	0.5
Meeting Minutes	1.0
Organization and Presentation	1.0
Project Demo (Two Basic Features)	1.0
Total	15 (equivalent to 3% course weight)

Appendix A – Sprint 1 Work Plan

Provide a Sprint 1 work plan that summarizes how your team intends to complete the Sprint 1 deliverables. The plan should be based on the issues created in your project repository and demonstrate that the required work has been identified, assigned, and scheduled.

For each planned item, include:

- Issue Number
- Issue Title
- Issue Type (User Story or Task)
- Responsible Member

- Target Completion Date
- Priority (High, Medium, Low)
- Current Status (Not Started, In Progress, Completed)

The purpose of this section is to provide a clear overview of the team's plan for Sprint 1. It should not contain detailed issue descriptions; instead, it should summarize the repository issues and show how the team has organized the work, assigned responsibilities, and planned to complete the sprint deliverables.

Appendix B- Example of user stories

US-01: As a job seeker, I want to create an account so that I can access platform features.

- *Task 1.1:* Design database schema for user account data.
- *Task 1.2:* Implement authentication API (signup endpoint).
- *Task 1.3:* Build user registration UI form.